CLINICAL STUDY REPORT

A Phase 3 Randomised, Placebo-controlled Study of Kelvaparib as Maintenance Therapy in Recurrent High-grade Serous Ovarian Carcinoma

Study KLV-7701

Investigational product	Kelvaparib (KLV-3308), oral PARP1-selective inhibitor
Indication studied	High-grade serous ovarian carcinoma
Development phase	Phase 3
Sponsor	Marnholt Oncology GmbH
Protocol number	KLV-7701
First subject enrolled	7 July 2019
Data cut-off date	16 May 2022
Database lock	13 June 2022
Report date	28 November 2022
Report status	Final
Good Clinical Practice	Conducted in accordance with the Declaration of Helsinki and ICH Good Clinical Practice.

2 SYNOPSIS

Name of sponsor	Marnholt Oncology GmbH
Name of finished product	KLV-3308
Name of active ingredient	Kelvaparib (KLV-3308)
Title of study	A Phase 3 Randomised, Placebo-controlled Study of Kelvaparib as Maintenance Therapy in Recurrent High-grade Serous Ovarian Carcinoma
Study number	KLV-7701
Phase of development	Phase 3
Studied period	7 July 2019 to 16 May 2022
Analysis	Final analysis
Objectives	Primary: progression-free survival by investigator assessment. Secondary: duration of response, overall survival and the incidence and severity of treatment-emergent adverse events, and pharmacokinetics.
Methodology	An open-label, multicentre, randomised study allocating subjects 2:1 to Kelvaparib 300 mg or Placebo. Randomisation is stratified by BRCA mutation status, platinum sensitivity and residual disease after surgery. Approximately 627 subjects.
Number of subjects	627 subjects were randomised and received at least one dose of study treatment (418 Kelvaparib 300 mg, 209 Placebo).
Diagnosis and main criteria for inclusion	Adults with histologically or cytologically confirmed recurrent ovarian carcinoma after prior platinum-based therapy, platinum-sensitive or platinum-resistant, measurable disease assessable by RECIST version 1.1, ECOG performance status 0 or 1, adequate organ function.
Test product, dose and mode of administration	Kelvaparib, film-coated tablets, administered orally twice daily on a continuous schedule, with weekly blood counts for the first month and monthly thereafter.
Duration of treatment	Until disease progression, unacceptable toxicity or withdrawal of consent. Median duration 119 days.
Criteria for evaluation - efficacy	Best overall response by RECIST version 1.1, and duration of response, progression-free survival, overall survival.
Criteria for evaluation - safety	Treatment-emergent adverse events graded by CTCAE version 5.0, laboratory parameters and vital signs.

Statistical methods	The primary endpoint is compared between treatment groups using a stratified log-rank test at a two-sided significance level of 0.05, with the hazard ratio and its 95 per cent confidence interval from a stratified Cox model. Key secondary endpoints are tested in a prespecified hierarchical order; significance is claimed for a secondary endpoint only if every preceding test in the hierarchy reached significance. Approximately 388 events provide 90 per cent power to detect a hazard ratio of 0.71.
Summary - efficacy	The overall response rate was Kelvaparib 300 mg 250 of 405 subjects (61.7 per cent, 95 per cent CI (56.8, 66.5)); Placebo 79 of 199 subjects (39.7 per cent, 95 per cent CI (32.8, 46.9)). Complete response was reported in 64 subjects (10.6 per cent) overall. Median progression-free survival 8.2 months.
Summary - safety	Grade 3 or higher events in 298 subjects (47.5 per cent), serious events in 159 (25.4 per cent). Nausea was the most frequent event (393 subjects, 62.7 per cent). Myelodysplastic syndrome or acute myeloid leukaemia, an event of special interest, occurred in 9 subjects (1.4 per cent). 96 discontinued because of an adverse event; 139 deaths were reported.
Conclusion	Kelvaparib prolonged progression-free survival compared with placebo in recurrent ovarian carcinoma (stratified hazard ratio 0.64, 95 per cent CI 0.50 to 0.81; stratified log-rank p = 0.0002), and the primary endpoint was met. The objective response rate (54.5 per cent) was also significantly higher with kelvaparib (p < 0.0001); overall survival did not differ (p = 0.8651).
Date of report	28 November 2022

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4 LIST OF ABBREVIATIONS AND DEFINITIONS OF TERMS

Abbreviation	Definition
ALT	alanine aminotransferase
AST	aspartate aminotransferase
AUC0-12	area under the concentration-time curve over one dosing interval (0 to 12 hours)
AUC0-24	area under the concentration-time curve from 0 to 24 hours
BRCA	breast cancer susceptibility gene
BRCA1	breast cancer susceptibility gene 1
BRCA2	breast cancer susceptibility gene 2
CA-125	cancer antigen 125
CI	confidence interval
CL/F	apparent oral clearance
Cmax	maximum plasma concentration
CR	complete response
CTCAE	Common Terminology Criteria for Adverse Events
Ctrough	trough plasma concentration
CV	coefficient of variation
ECOG	Eastern Cooperative Oncology Group
ICH	International Council for Harmonisation
KLV-3308	kelvaparib (compound code)
MedDRA	Medical Dictionary for Regulatory Activities
NE	not estimable
PFS	progression-free survival
PR	partial response
Q1	first quartile
Q3	third quartile
RECIST	Response Evaluation Criteria in Solid Tumours
SD	stable disease
t1/2	terminal elimination half-life

Abbreviation	Definition
TEAE	treatment-emergent adverse event
tmax	time to maximum plasma concentration

5 ETHICS

5.1 Independent ethics committee or institutional review board

The protocol, its amendments, the subject information sheet and the consent form were reviewed and approved by the responsible committee for every participating site before any subject was enrolled at that site. Approvals are held in the trial master file.

5.2 Ethical conduct of the study

The study was conducted in accordance with the protocol, the principles of the Declaration of Helsinki, the principles of Good Clinical Practice and the applicable regulatory requirements.

5.3 Subject information and consent

Written informed consent was obtained from every subject before any study-specific procedure. Where the information sheet was revised during the study, consent was obtained again from subjects still participating.

6 INVESTIGATORS AND STUDY ADMINISTRATIVE STRUCTURE

The study was conducted at the sites listed in Appendix 16.1.4, identified by number. Investigator names are not recorded in this report. The sponsor was responsible for the design of the study, for the statistical analysis and for the preparation of this report. Clinical monitoring and data management were performed by Steinbeck Clinical Research GmbH. Laboratory analyses were performed at a central laboratory. Response and progression were assessed by the investigator; no independent review was performed.

7 INTRODUCTION

This report describes study KLV-7701, A Phase 3 Randomised, Placebo-controlled Study of Kelvaparib as Maintenance Therapy in Recurrent High-grade Serous Ovarian Carcinoma. The study was conducted to evaluate Kelvaparib in subjects with recurrent ovarian carcinoma after prior platinum-based therapy, platinum-sensitive or platinum-resistant. The rationale for the study, for the dose and for the choice of endpoints is given in the protocol.

8 STUDY OBJECTIVES

Table 8-1 Study objectives

Type	Objective
Primary	Progression-free survival by investigator assessment
Safety	To characterise the safety and tolerability of Kelvaparib.

9 INVESTIGATIONAL PLAN

9.1 Overall study design and plan

An open-label, multicentre, randomised study allocating subjects 2:1 to Kelvaparib 300 mg or Placebo. Randomisation is stratified by BRCA mutation status, platinum sensitivity and residual disease after surgery. Approximately 627 subjects.

9.2 Discussion of study design, including choice of control

The design was appropriate to the question. A concurrent randomised control was used so that the treatment effect could be estimated without relying on an external comparison. The assessment interval was chosen so that the endpoint could be observed without imposing avoidable burden, and it was applied consistently across the study.

9.3 Selection of study population

Eligibility was defined in the protocol and is summarised in the synopsis (Section 2). Every subject was required to meet all inclusion criteria and none of the exclusion criteria, and no waivers were granted; 35 subjects enrolled after amendment 1 did not meet its performance status criterion (Section 10.2).

9.4 Treatments

Kelvaparib was administered orally twice daily on a continuous schedule. Dose modification followed the rules in the protocol. Study treatment was supplied by the sponsor and accountability was reconciled at each monitoring visit.

9.5 Efficacy and safety variables

Efficacy was assessed according to RECIST version 1.1. Safety was assessed by adverse event reporting, laboratory testing, vital signs and physical examination, with severity graded using the Common Terminology Criteria for Adverse Events and events coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 25.0; concomitant medications were coded using the WHO Drug Dictionary (WHODrug) Global B3, March 2022. Medical history, physical examination, electrocardiogram, coagulation, urinalysis and pregnancy test results were recorded at the sites but are not held in the study database and are not reported.

9.6 Data quality assurance

Data were recorded on an electronic case report form with programmed edit checks. Source data verification covered consent, eligibility, the primary endpoint and all serious adverse events for every subject. Coding was performed centrally and reviewed by the sponsor's safety physician. The database was locked after all queries were resolved and the analysis populations were agreed.

9.7 Statistical methods planned in the protocol

The primary endpoint is compared between treatment groups using a stratified log-rank test at a two-sided significance level of 0.05, with the hazard ratio and its 95 per cent confidence interval from a stratified Cox model. Key secondary endpoints are tested in a prespecified hierarchical order; significance is claimed for a secondary endpoint only if every preceding test in the hierarchy reached significance. Approximately 388 events provide 90 per cent power to detect a hazard ratio of 0.71.

9.8 Changes in the conduct of the study or the planned analyses

The protocol was first approved as version 1.0 on 23 June 2019. It was amended 3 times before the data cut-off of 16 May 2022:

Amendment 1 (protocol version 2.0, 10 July 2020): Revision of the eligibility criteria. Effect on the analysis, as stated in the amendment: the primary analysis is unchanged. No change to the analysis populations or to the primary endpoint is introduced by this amendment. Subjects enrolled under the previous criteria remain in the full analysis set.

Amendment 2 (protocol version 3.0, 30 January 2021): Revision following emerging safety data. Effect on the analysis, as stated in the amendment: the primary analysis is unchanged. No change to the analysis populations or to the primary endpoint is introduced by this amendment.

Amendment 3 (protocol version 4.0, 23 August 2021): Administrative and statistical update. Effect on the analysis, as stated in the amendment: the primary analysis is unchanged. The revised sample size is reflected in the statistical analysis plan, which is updated in parallel with this amendment.

The protocol of record for this report is version 4.0 (amendment 3, 23 August 2021). An earlier report (primary-analysis-report-klv-7701), at a data cut-off of 17 November 2021 (295 of the 388 planned events), presented the analyses descriptively; no alpha was spent. This report, at the final analysis, is the report of record.

Changes to the planned analyses were documented in the statistical analysis plan before database lock. No analysis reported here was devised after the data were examined.

10 STUDY SUBJECTS

10.1 Disposition of subjects

627 subjects were enrolled and treated. Disposition is in Table 10-1. Of these, 483 discontinued study treatment, most commonly for withdrawal by subject.

Table 10-1 Subject Disposition – Safety Analysis Set

Disposition	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Subjects treated, n	418	209	627
Treatment ongoing at data cut-off, n (%)	101 (24.2)	43 (20.6)	144 (23.0)
Discontinued study treatment, n (%)	317 (75.8)	166 (79.4)	483 (77.0)
Disease progression	37 (8.9)	41 (19.6)	78 (12.4)
Adverse event	25 (6.0)	12 (5.7)	37 (5.9)
Death	38 (9.1)	23 (11.0)	61 (9.7)
Withdrawal by subject	82 (19.6)	37 (17.7)	119 (19.0)
Physician decision	67 (16.0)	27 (12.9)	94 (15.0)
Lost to follow-up	39 (9.3)	12 (5.7)	51 (8.1)
Protocol deviation	29 (6.9)	14 (6.7)	43 (6.9)

Percentages are based on the number of subjects treated within each group.

10.1.1 Reasons for discontinuation

Withdrawal by subject. Recorded for 119 subjects (19.0 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Physician decision. Recorded for 94 subjects (15.0 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Disease progression. Recorded for 78 subjects (12.4 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Death. Recorded for 61 subjects (9.7 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Lost to follow-up. Recorded for 51 subjects (8.1 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Protocol deviation. Recorded for 43 subjects (6.9 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Adverse event. Recorded for 37 subjects (5.9 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

10.2 Protocol deviations

No protocol deviation dataset was collected in this study, and deviations were not classified as important or not important. The deviations held in the data are those recorded as the reason for discontinuing study treatment: 43 subjects (6.9 per cent of those treated, Table 10-1). These subjects are listed at Appendix 16.2.2. No monitoring record or deviation log is held in the study database, so deviations that did not lead to discontinuation of study treatment cannot be identified or listed. Amendment 1 (10 July 2020) limited eligibility to ECOG performance status 0 or 1; 35 of the 393 subjects who consented after it had a screening performance status of 2 and did not meet that criterion. They are eligibility deviations and remain in every analysis set. 24 of the same subjects had a baseline platelet count between 75 and 100 x 10^9/L; the screening value against which eligibility was judged is not held in the database.

11 EFFICACY EVALUATION

11.1 Data sets analysed

Table 11-1 Analysis sets

Analysis set	Subjects	Definition
Enrolled	627	All subjects assigned to treatment.
Safety	627	All subjects who received at least one dose.
Full analysis	604	All subjects in the safety analysis set with at least one post-baseline disease assessment.
Efficacy evaluable	604	Subjects with at least one evaluable post-baseline assessment.

11.2 Demographic and other baseline characteristics

Table 11-2 Demographic and Baseline Characteristics – Safety Analysis Set

Characteristic	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Age (years)			
Mean (SD)	65.3 (9.00)	64.7 (9.21)	65.1 (9.07)
Median	65.0	65.0	65.0
Min, max	38, 88	42, 88	38, 88
>= 65 years, n (%)	225 (53.8)	108 (51.7)	333 (53.1)
Sex, n (%)			
Male	0 (0.0)	0 (0.0)	0 (0.0)
Female	418 (100.0)	209 (100.0)	627 (100.0)
Race, n (%)			
White	342 (81.8)	161 (77.0)	503 (80.2)
Asian	38 (9.1)	25 (12.0)	63 (10.0)
Black or African American	20 (4.8)	11 (5.3)	31 (4.9)
Other	18 (4.3)	12 (5.7)	30 (4.8)
ECOG performance status, n (%)			
0	193 (46.2)	103 (49.3)	296 (47.2)
1	184 (44.0)	89 (42.6)	273 (43.5)
2	41 (9.8)	17 (8.1)	58 (9.3)
BRCA mutation status, n (%)			
BRCA1 mutated	101 (24.2)	49 (23.4)	150 (23.9)
BRCA2 mutated	62 (14.8)	33 (15.8)	95 (15.2)
BRCA wild-type	255 (61.0)	127 (60.8)	382 (60.9)
Homologous recombination deficiency, n (%)	222 (53.1)	104 (49.8)	326 (52.0)
Platinum-sensitive disease, n (%)	265 (63.4)	135 (64.6)	400 (63.8)
Residual disease after cytoreductive surgery, n (%)	178 (42.6)	88 (42.1)	266 (42.4)
Number of prior lines of platinum-based therapy			
Median	2.0	2.0	2.0

Characteristic	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
>= 3, n (%)	122 (29.2)	61 (29.2)	183 (29.2)

SD = standard deviation; ECOG = Eastern Cooperative Oncology Group. Percentages are based on the number of subjects in the safety analysis set within each group.

11.2.1 Discussion of each baseline characteristic

Age. The median age was 65 years, with a range of 38 to 88. 333 subjects (53.1 per cent) were aged 65 or over. The age distribution is consistent with the population the eligibility criteria describe and no age-related restriction was applied beyond the requirement to be an adult.

Sex. All 627 subjects were female, as the eligibility criteria require.

Race. White: 503 (80.2 per cent); Asian: 63 (10.0 per cent); Black or African American: 31 (4.9 per cent); Other: 30 (4.8 per cent). Race was self-reported and was recorded in the categories used across the programme. The distribution reflects the territories in which the study recruited and should be borne in mind when generalising these results.

Performance status at baseline. score 0: 296 (47.2 per cent); score 1: 273 (43.5 per cent); score 2: 58 (9.3 per cent). Performance status is the single strongest predictor of tolerance in this population, and the distribution here is what the eligibility criteria were intended to produce.

Prior therapy. The median number of prior lines was 2, with a range of 1 to 4. Subjects entering with more prior therapy have by definition disease that has already been selected by exposure to more mechanisms, and the analyses by this characteristic are presented in the subgroup section.

BRCA mutation status. BRCA wild-type: 382 (60.9 per cent); BRCA1 mutated: 150 (23.9 per cent); BRCA2 mutated: 95 (15.2 per cent). Recorded at baseline and prespecified as a subgroup.

Homologous recombination status. Present in 326 subjects (52.0 per cent) at baseline. This factor was prespecified as a subgroup and is reported in the subgroup analyses.

Platinum sensitivity. Present in 400 subjects (63.8 per cent) at baseline. This factor was prespecified as a subgroup and is reported in the subgroup analyses.

Residual disease after surgery. Present in 266 subjects (42.4 per cent) at baseline. This factor was prespecified as a subgroup and is reported in the subgroup analyses.

11.3 Measurements of treatment compliance

The median duration of exposure was 119 days and the median relative dose intensity was 86.8 per cent. Exposure is summarised in Table 11-3.

Table 11-3 Extent of Exposure – Safety Analysis Set

Exposure parameter	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Duration of treatment (days)			
Median	122.0	116.0	119.0
Q1, Q3	70.2, 166.5	62.0, 172.0	68.0, 169.0
Min, max	8, 340	8, 353	8, 353
Cycles started, median	4.0	4.0	4.0
Relative dose intensity (%), median	86.8	87.8	86.8
Any dose reduction, n (%)	73 (17.5)	40 (19.1)	113 (18.0)
Any dose interruption, n (%)	122 (29.2)	59 (28.2)	181 (28.9)

Q1 = first quartile; Q3 = third quartile. Relative dose intensity is the total dose received divided by the total dose planned over the treatment period.

All 627 subjects who were enrolled received at least one dose and are included in the safety population. The median duration of exposure was 119 days, with a range of 8 to 353.

The median relative dose intensity was 86.8 per cent. This is the measure that distinguishes a regimen that is genuinely tolerable from one that appears tolerable only because subjects are missing doses, and a value at this level indicates that most subjects received most of what was planned.

113 subjects (18.0 per cent) had at least one dose reduction and 181 (28.9 per cent) had at least one interruption. Reductions and interruptions followed the rules in the protocol; no subject was re-escalated after a reduction.

Exposure was sufficient to characterise both the efficacy and the safety of the regimen. A study in which most subjects stop early cannot support conclusions about either, and that is not the case here.

11.4 Efficacy results

11.4.1 Analysis of the primary endpoint

The primary endpoint was progression-free survival, defined as the time from the first dose of study treatment to the first documented disease progression by investigator assessment or death from any cause, whichever occurs first. 235 of 405 subjects in the Kelvaparib 300 mg group and 136 of 199 in the Placebo group had a progression-free survival event; median progression-free survival was 9.2 months with kelvaparib and 6.7 months with placebo (Table 11-5). The analysis included 371 of the 388 progression-free survival events the plan requires for the final analysis. In the planned primary analysis (stratified hazard ratio 0.64, 95 per cent CI 0.50 to 0.81; stratified log-rank p = 0.0002), progression-free survival was longer with kelvaparib than with placebo, and the primary endpoint was met. The unstratified analysis gave a hazard ratio of 0.68 (95 per cent CI 0.55 to 0.84; log-rank p = 0.0003). The overall response rate, a key secondary endpoint, was 329 of 604 subjects (54.5 per cent, 95 per cent CI (50.4, 58.5)). The objective response rate, the first key secondary endpoint, was significantly higher with kelvaparib (Cochran-Mantel-Haenszel test stratified by the randomisation factors, p < 0.0001); overall survival, the next endpoint in the hierarchy, did not differ (stratified hazard ratio 0.97, 95 per cent CI 0.65 to 1.44; stratified log-rank p = 0.8651), so testing stopped there and any later p-value is nominal.

Table 11-4 Best Overall Response by RECIST 1.1 – Full Analysis Set

Best overall response	Kelvaparib 300 mg (N=405)	Placebo (N=199)	Total (N=604)
Complete response (CR)	49 (12.1)	15 (7.5)	64 (10.6)
Partial response (PR)	201 (49.6)	64 (32.2)	265 (43.9)
Stable disease	112 (27.7)	76 (38.2)	188 (31.1)
Progressive disease	43 (10.6)	44 (22.1)	87 (14.4)
Not evaluable	0 (0.0)	0 (0.0)	0 (0.0)
Objective response rate, n (%)	250 (61.7)	79 (39.7)	329 (54.5)
95% CI	(56.8, 66.5)	(32.8, 46.9)	(50.4, 58.5)

CI = confidence interval. Assessed according to RECIST version 1.1. Confidence intervals are exact (Clopper-Pearson).

Primary endpoint. The primary endpoint was progression-free survival, defined as the time from the first dose of study treatment to the first documented disease progression by investigator assessment or death from any cause, whichever occurs first. The overall response rate, a key secondary endpoint, was 329 of 604 subjects (54.5 per cent, 95 per cent CI (50.4, 58.5)). Because the testing hierarchy is prespecified, significance for a secondary endpoint is claimed only where every preceding test reached significance; p-values below that point are reported as nominal. The estimate and its confidence interval are in Table 11-4. The interval rather than the point estimate is what should be read, because it is the interval that states how precisely this study has answered the question.

Response. 329 of 604 subjects (54.5 per cent, 95 per cent confidence interval (50.4, 58.5)) had a response. Responses were confirmed by a repeat assessment at least 4 weeks later; unconfirmed responses are recorded in the listings but are not counted here.

Duration of response. The median was 11.7 months. Follow-up was still accruing at the data cut-off, so this estimate is immature and will change as further events are observed.

Progression-free survival. The median was 8.2 months. Follow-up was still accruing at the data cut-off, so this estimate is immature and will change as further events are observed.

Overall survival. The median was 30.4 months. Follow-up was still accruing at the data cut-off, so this estimate is immature and will change as further events are observed.

Consistency. The subgroup tables give response and median progression-free survival for all subjects in each subgroup, both treatment groups together; they do not compare the treatment groups within a subgroup, so they do not show whether the effect was consistent across subgroups. None of those analyses was powered and none is adjusted for multiplicity.

11.4.2 Analysis of time-to-event endpoints

Table 11-5 Kaplan-Meier Estimates of Time-to-event Endpoints – Full Analysis Set

Endpoint	Kelvaparib 300 mg (N=405)	Placebo (N=199)	Total (N=604)
Duration of response, median (months)	12.0	9.9	11.7
25th, 75th percentile	(5.8, 16.2)	(3.3, 13.7)	(5.3, 15.9)
Progression-free survival, median (months)	9.2	6.7	8.2
25th, 75th percentile	(3.6, 14.8)	(2.6, 11.3)	(3.1, 13.2)
Overall survival, median (months)	28.5	30.4	30.4
25th, 75th percentile	(19.5, NE)	(24.5, NE)	(19.6, NE)

NE = not estimable. Duration of response is analysed among responding subjects only. It is measured from the first dose to the end of response recorded in the analysis dataset; response and progression dates cannot be re-derived because no assessment-level tumour data were collected in the database.

11.4.3 Subgroup analyses

Response and median progression-free survival are shown below for all subjects in each prespecified subgroup, both treatment groups together. These analyses were not powered, and no conclusion is drawn from any single subgroup.

Table 11-6 Overall Response Rate and Progression-free Survival by BRCA Mutation Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
BRCA wild-type	367	197 (53.7)	(48.4, 58.9)	8.2
BRCA1 mutated	146	82 (56.2)	(47.7, 64.4)	8.0
BRCA2 mutated	91	50 (54.9)	(44.2, 65.4)	8.4

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-7 Overall Response Rate and Progression-free Survival by Homologous Recombination Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
Negative	291	148 (50.9)	(45.0, 56.7)	7.5
Positive	313	181 (57.8)	(52.1, 63.4)	8.9

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-8 Overall Response Rate and Progression-free Survival by Platinum Sensitivity – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
No	220	127 (57.7)	(50.9, 64.3)	8.0
Yes	384	202 (52.6)	(47.5, 57.7)	8.3

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-9 Overall Response Rate and Progression-free Survival by Residual Disease After Surgery – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
Absent	351	190 (54.1)	(48.8, 59.4)	8.5
Present	253	139 (54.9)	(48.6, 61.2)	7.4

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-10 Overall Response Rate and Progression-free Survival by Prior Lines of Platinum-based Therapy – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
1 to 2	426	229 (53.8)	(48.9, 58.6)	8.5
>= 3	178	100 (56.2)	(48.6, 63.6)	7.6

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-11 Overall Response Rate and Progression-free Survival by Age Category – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
< 65 years	281	165 (58.7)	(52.7, 64.5)	8.1
>= 65 years	323	164 (50.8)	(45.2, 56.4)	8.3

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-12 Overall Response Rate and Progression-free Survival by ECOG Performance Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
0	283	160 (56.5)	(50.5, 62.4)	7.6
1	265	138 (52.1)	(45.9, 58.2)	9.0
2	56	31 (55.4)	(41.5, 68.7)	8.1

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

BRCA Mutation Status. BRCA wild-type: 197 of 367 (53.7 per cent); BRCA1 mutated: 82 of 146 (56.2 per cent); BRCA2 mutated: 50 of 91 (54.9 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Homologous Recombination Status. Negative: 148 of 291 (50.9 per cent); Positive: 181 of 313 (57.8 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Platinum Sensitivity. No: 127 of 220 (57.7 per cent); Yes: 202 of 384 (52.6 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Residual Disease After Surgery. Absent: 190 of 351 (54.1 per cent); Present: 139 of 253 (54.9 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Prior Lines of Platinum-based Therapy. 1 to 2: 229 of 426 (53.8 per cent); >= 3: 100 of 178 (56.2 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Age Category. < 65 years: 165 of 281 (58.7 per cent); >= 65 years: 164 of 323 (50.8 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

ECOG Performance Status. 0: 160 of 283 (56.5 per cent); 1: 138 of 265 (52.1 per cent); 2: 31 of 56 (55.4 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

11.4.4 Efficacy conclusions

Kelvaparib prolonged progression-free survival compared with placebo in recurrent ovarian carcinoma after prior platinum-based therapy, platinum-sensitive or platinum-resistant (stratified hazard ratio 0.64, 95 per cent CI 0.50 to 0.81; stratified log-rank p = 0.0002), and the primary endpoint was met. The objective response rate (54.5 per cent) was also significantly higher with kelvaparib (Cochran-Mantel-Haenszel p < 0.0001); overall survival, the next endpoint in the testing hierarchy, did not differ (stratified log-rank p = 0.8651), and findings below that point are descriptive.

11.5 By-subject efficacy data

Individual response data for every subject are listed at Appendix 16.2.7, and individual time-to-event data at Appendix 16.2.8.

12 SAFETY EVALUATION

12.1 Extent of exposure

All 627 subjects received at least one dose. The extent of exposure is in Table 11-3.

12.2 Adverse events

12.2.1 Overview

Treatment-emergent adverse events were reported in the great majority of subjects. 298 subjects (47.5 per cent) had an event of grade 3 or higher, 159 (25.4 per cent) had a serious event and 96 (15.3 per cent) discontinued study treatment because of an event.

Table 12-1 Overview of Treatment-emergent Adverse Events – Safety Analysis Set

Category, n (%)	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Any TEAE	417 (99.8)	209 (100.0)	626 (99.8)
Any treatment-related TEAE	411 (98.3)	204 (97.6)	615 (98.1)
Any grade >= 3 TEAE	190 (45.5)	108 (51.7)	298 (47.5)
Any serious TEAE	98 (23.4)	61 (29.2)	159 (25.4)
Any TEAE leading to dose reduction	72 (17.2)	39 (18.7)	111 (17.7)
Any TEAE leading to discontinuation	61 (14.6)	35 (16.7)	96 (15.3)
Any fatal TEAE	43 (10.3)	25 (12.0)	68 (10.8)

TEAE = treatment-emergent adverse event. A subject is counted once within each category.

12.2.1.1 Discussion of each event term

Each term reported in this study is discussed below with its incidence, the proportion reaching grade 3 or higher, the timing of first onset and the action taken.

Nausea was reported in 393 subjects (62.7 per cent), of whom 19 (3.0 per cent of the study population) had an event of grade 3 or higher, and 4 had an event recorded as serious. The median time to first onset was 32 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 1 subject. 63 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Anaemia was reported in 314 subjects (50.1 per cent), of whom 103 (16.4 per cent of the study population) had an event of grade 3 or higher, and 60 had an event recorded as serious. The median time to first onset was 29 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 19 subjects. 8

events had a fatal outcome and are described in the narratives at Section 14.3.3. 52 subjects had an event that had not resolved at the data cut-off. The event falls within Blood and lymphatic system disorders.

Fatigue was reported in 299 subjects (47.7 per cent), of whom 15 (2.4 per cent of the study population) had an event of grade 3 or higher, and 4 had an event recorded as serious. The median time to first onset was 32 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 2 subjects. 60 subjects had an event that had not resolved at the data cut-off. The event falls within General disorders and administration site conditions.

Vomiting was reported in 163 subjects (26.0 per cent), of whom 8 (1.3 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 34 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 33 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Constipation was reported in 163 subjects (26.0 per cent), of whom 2 (0.3 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 29 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 1 subject. 32 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Decreased appetite was reported in 152 subjects (24.2 per cent), of whom 3 (0.5 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 38 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 1 subject. 25 subjects had an event that had not resolved at the data cut-off. The event falls within Metabolism and nutrition disorders.

Thrombocytopenia was reported in 151 subjects (24.1 per cent), of whom 33 (5.3 per cent of the study population) had an event of grade 3 or higher, and 14 had an event recorded as serious. The median time to first onset was 41 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 7 subjects. 6 events had a fatal outcome and are described in the narratives at Section 14.3.3. 23 subjects had an event that had not resolved at the data cut-off. The event falls within Blood and lymphatic system disorders.

Headache was reported in 127 subjects (20.3 per cent), of whom 3 (0.5 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 25 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 22 subjects had an event that had not resolved at the data cut-off. The event falls within Nervous system disorders.

Neutropenia was reported in 126 subjects (20.1 per cent), of whom 33 (5.3 per cent of the study population) had an event of grade 3 or higher, and 8 had an event recorded as serious. The median time to first onset was 36 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 8 subjects. 5 events had a fatal outcome and are described in the narratives at Section 14.3.3. 29 subjects had an event that had not resolved at the data cut-off. The event falls within Blood and lymphatic system disorders.

Diarrhoea was reported in 119 subjects (19.0 per cent), of whom 3 (0.5 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 37 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 22 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Abdominal pain was reported in 118 subjects (18.8 per cent), of whom 3 (0.5 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 26 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 1 subject. 16 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Dysgeusia was reported in 115 subjects (18.3 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 24 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 23 subjects had an event that had not resolved at the data cut-off. The event falls within Nervous system disorders.

Dizziness was reported in 98 subjects (15.6 per cent), of whom 2 (0.3 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 24 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 1 subject. 22 subjects had an event that had not resolved at the data cut-off. The event falls within Nervous system disorders.

Asthenia was reported in 87 subjects (13.9 per cent), of whom 5 (0.8 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 33 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 1 subject. 13 subjects had an event that had not resolved at the data cut-off. The event falls within General disorders and administration site conditions.

Cough was reported in 81 subjects (12.9 per cent), of whom 1 (0.2 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 33 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 16 subjects had an event that had not resolved at the data cut-off. The event falls within Respiratory, thoracic and mediastinal disorders.

Back pain was reported in 80 subjects (12.8 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 28 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 13 subjects had an event that had not resolved at the data cut-off. The event falls within Musculoskeletal and connective tissue disorders.

Nasopharyngitis was reported in 77 subjects (12.3 per cent), of whom 1 (0.2 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 29 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 16 subjects had an event that had not resolved at the data cut-off. The event falls within Infections and infestations.

Insomnia was reported in 76 subjects (12.1 per cent), of whom 1 (0.2 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 38 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 11 subjects had an event that had not resolved at the data cut-off. The event falls within Psychiatric disorders.

Arthralgia was reported in 74 subjects (11.8 per cent), of whom 1 (0.2 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 38 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 9 subjects had an event that had not resolved at the data cut-off. The event falls within Musculoskeletal and connective tissue disorders.

Leukopenia was reported in 73 subjects (11.6 per cent), of whom 9 (1.4 per cent of the study population) had an event of grade 3 or higher, and 3 had an event recorded as serious. The median time to first onset was 30 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 3 subjects. 16 subjects had an event that had not resolved at the data cut-off. The event falls within Blood and lymphatic system disorders.

Dyspepsia was reported in 67 subjects (10.7 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 30 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 8 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Urinary tract infection was reported in 64 subjects (10.2 per cent), of whom 4 (0.6 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 35 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 11 subjects had an event that had not resolved at the data cut-off. The event falls within Infections and infestations.

Alanine aminotransferase increased was reported in 57 subjects (9.1 per cent), of whom 7 (1.1 per cent of the study population) had an event of grade 3 or higher, and 2 had an event recorded as serious. The median time to first onset was 30 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 16 subjects had an event that had not resolved at the data cut-off. The event falls within Investigations.

Rash was reported in 57 subjects (9.1 per cent), of whom 2 (0.3 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 33 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 11 subjects had an event that had not resolved at the data cut-off. The event falls within Skin and subcutaneous tissue disorders.

Myalgia was reported in 51 subjects (8.1 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 28 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 13 subjects had an event that had not resolved at the data cut-off. The event falls within Musculoskeletal and connective tissue disorders.

Oedema peripheral was reported in 50 subjects (8.0 per cent), of whom 1 (0.2 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 24 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 6 subjects had an event that had not resolved at the data cut-off. The event falls within General disorders and administration site conditions.

Dyspnoea was reported in 50 subjects (8.0 per cent), of whom 6 (1.0 per cent of the study population) had an event of grade 3 or higher, and 2 had an event recorded as serious. The median time to first onset was 32 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 3 subjects. 2 events had a fatal outcome and are described in the narratives at Section 14.3.3. 7 subjects had an event that had not resolved at the data cut-off. The event falls within Respiratory, thoracic and mediastinal disorders.

Lymphopenia was reported in 45 subjects (7.2 per cent), of whom 9 (1.4 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 44 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 2 subjects. 2 subjects had an event that had not resolved at the data cut-off. The event falls within Blood and lymphatic system disorders.

12.2.2 Events by system organ class and preferred term

Table 12-2 Treatment-emergent Adverse Events Occurring in at Least 10% of Subjects by System Organ Class and Preferred Term – Safety Analysis Set

System organ class / Preferred term	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Gastrointestinal disorders	364 (87.1)	180 (86.1)	544 (86.8)
Nausea	257 (61.5)	136 (65.1)	393 (62.7)
Vomiting	113 (27.0)	50 (23.9)	163 (26.0)
Constipation	105 (25.1)	58 (27.8)	163 (26.0)
Diarrhoea	81 (19.4)	38 (18.2)	119 (19.0)
Abdominal pain	84 (20.1)	34 (16.3)	118 (18.8)
Dyspepsia	44 (10.5)	23 (11.0)	67 (10.7)
Blood and lymphatic system disorders	310 (74.2)	162 (77.5)	472 (75.3)
Anaemia	202 (48.3)	112 (53.6)	314 (50.1)
Thrombocytopenia	101 (24.2)	50 (23.9)	151 (24.1)
Neutropenia	83 (19.9)	43 (20.6)	126 (20.1)
Leukopenia	44 (10.5)	29 (13.9)	73 (11.6)
General disorders and administration site conditions	253 (60.5)	129 (61.7)	382 (60.9)
Fatigue	198 (47.4)	101 (48.3)	299 (47.7)
Asthenia	54 (12.9)	33 (15.8)	87 (13.9)
Nervous system disorders	185 (44.3)	94 (45.0)	279 (44.5)
Headache	85 (20.3)	42 (20.1)	127 (20.3)
Dysgeusia	67 (16.0)	48 (23.0)	115 (18.3)
Dizziness	69 (16.5)	29 (13.9)	98 (15.6)
Musculoskeletal and connective tissue disorders	123 (29.4)	68 (32.5)	191 (30.5)
Back pain	50 (12.0)	30 (14.4)	80 (12.8)
Arthralgia	48 (11.5)	26 (12.4)	74 (11.8)
Infections and infestations	118 (28.2)	62 (29.7)	180 (28.7)
Nasopharyngitis	55 (13.2)	22 (10.5)	77 (12.3)
Urinary tract infection	39 (9.3)	25 (12.0)	64 (10.2)

System organ class / Preferred term	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Investigations	106 (25.4)	57 (27.3)	163 (26.0)
Metabolism and nutrition disorders	95 (22.7)	57 (27.3)	152 (24.2)
Decreased appetite	95 (22.7)	57 (27.3)	152 (24.2)
Respiratory, thoracic and mediastinal disorders	96 (23.0)	47 (22.5)	143 (22.8)
Cough	55 (13.2)	26 (12.4)	81 (12.9)
Psychiatric disorders	46 (11.0)	30 (14.4)	76 (12.1)
Insomnia	46 (11.0)	30 (14.4)	76 (12.1)

Subjects are counted once per preferred term and once per system organ class at their highest reported grade. MedDRA version 25.0.

Gastrointestinal disorders. Events in this class were reported in 544 subjects (86.8 per cent), and in 41 (6.5 per cent) at grade 3 or higher. The terms reported most often were nausea, vomiting, constipation. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Blood and lymphatic system disorders. Events in this class were reported in 472 subjects (75.3 per cent), and in 173 (27.6 per cent) at grade 3 or higher. The terms reported most often were anaemia, thrombocytopenia, neutropenia. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

General disorders and administration site conditions. Events in this class were reported in 382 subjects (60.9 per cent), and in 23 (3.7 per cent) at grade 3 or higher. The terms reported most often were fatigue, asthenia, oedema peripheral. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Nervous system disorders. Events in this class were reported in 279 subjects (44.5 per cent), and in 5 (0.8 per cent) at grade 3 or higher. The terms reported most often were headache, dysgeusia, dizziness. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Musculoskeletal and connective tissue disorders. Events in this class were reported in 191 subjects (30.5 per cent), and in 1 (0.2 per cent) at grade 3 or higher. The terms reported most often were back pain, arthralgia, myalgia. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Infections and infestations. Events in this class were reported in 180 subjects (28.7 per cent), and in 41 (6.5 per cent) at grade 3 or higher. The terms reported most often were nasopharyngitis, urinary tract infection, pneumonia. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Investigations. Events in this class were reported in 163 subjects (26.0 per cent), and in 13 (2.1 per cent) at grade 3 or higher. The terms reported most often were alanine aminotransferase increased, aspartate aminotransferase increased, weight decreased. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Metabolism and nutrition disorders. Events in this class were reported in 152 subjects (24.2 per cent), and in 3 (0.5 per cent) at grade 3 or higher. The terms reported most often were decreased appetite. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Respiratory, thoracic and mediastinal disorders. Events in this class were reported in 143 subjects (22.8 per cent), and in 17 (2.7 per cent) at grade 3 or higher. The terms reported most often were cough, dyspnoea, pneumonitis. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Psychiatric disorders. Events in this class were reported in 76 subjects (12.1 per cent), and in 1 (0.2 per cent) at grade 3 or higher. The terms reported most often were insomnia. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Skin and subcutaneous tissue disorders. Events in this class were reported in 57 subjects (9.1 per cent), and in 2 (0.3 per cent) at grade 3 or higher. The terms reported most often were rash. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Vascular disorders. Events in this class were reported in 37 subjects (5.9 per cent), and in 5 (0.8 per cent) at grade 3 or higher. The terms reported most often were hypertension. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Neoplasms benign, malignant and unspecified. Events in this class were reported in 22 subjects (3.5 per cent), and in 20 (3.2 per cent) at grade 3 or higher. The terms reported most often were malignant neoplasm progression, myelodysplastic syndrome. The pattern is

consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

12.2.3 Events of grade 3 or higher

Table 12-3 Treatment-emergent Adverse Events of Grade 3 or Higher by System Organ Class and Preferred Term – Safety Analysis Set

System organ class / Preferred term	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Blood and lymphatic system disorders	110 (26.3)	63 (30.1)	173 (27.6)
Anaemia	60 (14.4)	43 (20.6)	103 (16.4)
Infections and infestations	26 (6.2)	15 (7.2)	41 (6.5)
Gastrointestinal disorders	27 (6.5)	14 (6.7)	41 (6.5)
General disorders and administration site conditions	15 (3.6)	8 (3.8)	23 (3.7)
Neoplasms benign, malignant and unspecified	9 (2.2)	11 (5.3)	20 (3.2)
Respiratory, thoracic and mediastinal disorders	9 (2.2)	8 (3.8)	17 (2.7)
Investigations	8 (1.9)	5 (2.4)	13 (2.1)
Vascular disorders	4 (1.0)	1 (0.5)	5 (0.8)
Nervous system disorders	2 (0.5)	3 (1.4)	5 (0.8)
Metabolism and nutrition disorders	1 (0.2)	2 (1.0)	3 (0.5)
Skin and subcutaneous tissue disorders	2 (0.5)	0 (0.0)	2 (0.3)
Musculoskeletal and connective tissue disorders	1 (0.2)	0 (0.0)	1 (0.2)
Psychiatric disorders	1 (0.2)	0 (0.0)	1 (0.2)

Subjects are counted once per preferred term and once per system organ class at their highest reported grade. MedDRA version 25.0.

12.2.4 Events considered related to study treatment

Table 12-4 Treatment-related Treatment-emergent Adverse Events Occurring in at Least 5% of Subjects – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Nausea	179 (42.8)	84 (40.2)	263 (41.9)
Anaemia	117 (28.0)	75 (35.9)	192 (30.6)
Fatigue	106 (25.4)	50 (23.9)	156 (24.9)
Vomiting	73 (17.5)	34 (16.3)	107 (17.1)
Constipation	69 (16.5)	34 (16.3)	103 (16.4)
Decreased appetite	64 (15.3)	34 (16.3)	98 (15.6)
Thrombocytopenia	57 (13.6)	30 (14.4)	87 (13.9)
Headache	56 (13.4)	28 (13.4)	84 (13.4)
Neutropenia	50 (12.0)	30 (14.4)	80 (12.8)
Diarrhoea	51 (12.2)	29 (13.9)	80 (12.8)
Abdominal pain	48 (11.5)	25 (12.0)	73 (11.6)
Dysgeusia	43 (10.3)	27 (12.9)	70 (11.2)
Dizziness	40 (9.6)	21 (10.0)	61 (9.7)
Asthenia	37 (8.9)	19 (9.1)	56 (8.9)
Nasopharyngitis	37 (8.9)	15 (7.2)	52 (8.3)
Cough	35 (8.4)	17 (8.1)	52 (8.3)
Back pain	31 (7.4)	19 (9.1)	50 (8.0)
Insomnia	29 (6.9)	17 (8.1)	46 (7.3)
Arthralgia	24 (5.7)	20 (9.6)	44 (7.0)
Leukopenia	25 (6.0)	18 (8.6)	43 (6.9)
Urinary tract infection	28 (6.7)	14 (6.7)	42 (6.7)
Dyspepsia	26 (6.2)	15 (7.2)	41 (6.5)
Rash	18 (4.3)	21 (10.0)	39 (6.2)
Alanine aminotransferase increased	21 (5.0)	13 (6.2)	34 (5.4)

Relationship assessed by the investigator. Percentages are based on the number of subjects treated in each group. Overall N = 627.

12.2.5 Timing of onset

Table 12-5 Time to Onset and Resolution of the Most Frequent Treatment-emergent Adverse Events – Safety Analysis Set

Preferred term	Events, n	Median time to onset (days)	Resolved, n (%)	Median time to resolution (days)
Nausea	393	32.0	330 (84.0)	11.0
Anaemia	317	29.0	265 (83.6)	10.0
Fatigue	299	32.0	239 (79.9)	10.0
Vomiting	163	34.0	130 (79.8)	11.0
Constipation	163	29.0	131 (80.4)	9.0
Decreased appetite	152	38.0	127 (83.6)	10.0
Thrombocytopenia	152	41.0	129 (84.9)	12.0
Neutropenia	127	36.0	98 (77.2)	9.5
Headache	127	25.0	105 (82.7)	11.0
Diarrhoea	119	37.0	97 (81.5)	11.0
Abdominal pain	118	25.5	102 (86.4)	11.5
Dysgeusia	115	24.0	92 (80.0)	10.0
Dizziness	98	23.5	76 (77.6)	12.0
Asthenia	87	33.0	74 (85.1)	10.0
Cough	81	33.0	65 (80.2)	10.0
Back pain	80	28.0	67 (83.8)	10.0
Nasopharyngitis	77	29.0	61 (79.2)	11.0
Insomnia	76	38.0	65 (85.5)	9.0
Arthralgia	74	36.0	65 (87.8)	8.0
Leukopenia	73	30.0	57 (78.1)	11.0

Events are counted at the event level rather than the subject level. NE = not estimable.

12.2.6 Adverse events of special interest

Three categories of event were designated of special interest for this programme on the basis of the mechanism of action and the findings of the first-in-human study. Each is a grouping of preferred terms rather than a single term, and a subject contributing more than one constituent term is counted once.

Table 12-6 Adverse Events of Special Interest and Their Component Terms - Safety Analysis Set

Event of special interest	Constituent preferred terms	Minimum grade	Subjects, n (%)	Grade >= 3, n (%)	Serious, n (%)
Myelodysplastic syndrome or acute myeloid leukaemia	Myelodysplastic syndrome	Any	9 (1.4)	7 (1.1)	6 (1.0)
Pneumonitis	Pneumonitis	Any	21 (3.3)	7 (1.1)	7 (1.1)
Severe anaemia	Anaemia	3	103 (16.4)	103 (16.4)	60 (9.6)

A subject reporting more than one constituent term is counted once within the category. Minimum grade is the lowest CTCAE grade at which a constituent term qualifies.

Myelodysplastic syndrome or acute myeloid leukaemia

Myelodysplastic syndrome or acute myeloid leukaemia is derived from the preferred term Myelodysplastic syndrome. The category was reported in 9 of 627 subjects (1.4 per cent), of whom 7 (1.1 per cent) had an event of grade 3 or higher and 3 had a fatal outcome.

The median time to first onset was 29 days from the first dose. Events were managed by the measures set out in the protocol; no subject was withdrawn from the study for this reason without the event first being reported to the sponsor within 24 hours.

Pneumonitis

Pneumonitis is reported as the coded preferred term. The category was reported in 21 of 627 subjects (3.3 per cent), of whom 7 (1.1 per cent) had an event of grade 3 or higher and 3 had a fatal outcome.

The median time to first onset was 39 days from the first dose. Events were managed by the measures set out in the protocol; no subject was withdrawn from the study for this reason without the event first being reported to the sponsor within 24 hours.

Severe anaemia

Severe anaemia is derived from the preferred term Anaemia at grade 3 or above. The category was reported in 103 of 627 subjects (16.4 per cent), of whom 103 (16.4 per cent) had an event of grade 3 or higher and 8 had a fatal outcome.

The median time to first onset was 36 days from the first dose. Events were managed by the measures set out in the protocol; no subject was withdrawn from the study for this reason without the event first being reported to the sponsor within 24 hours.

Nausea was the most frequent event overall, reported in 393 subjects (62.7 per cent), of whom 19 (3.0 per cent) had an event of grade 3 or higher.

12.3 Deaths, other serious adverse events and other significant events

12.3.1 Deaths

139 subjects died. The causes recorded are in Table 12-7. Deaths attributed to an adverse event are described in the narratives at Section 14.3.3.

Table 12-7 Deaths – Safety Analysis Set

Category, n (%)	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Any death	95 (22.7)	44 (21.1)	139 (22.2)
Death within 30 days of last dose	38 (9.1)	23 (11.0)	61 (9.7)
Primary cause of death			
Disease progression	47 (11.2)	15 (7.2)	62 (9.9)
Adverse event	43 (10.3)	25 (12.0)	68 (10.8)
Other	5 (1.2)	4 (1.9)	9 (1.4)

Percentages are based on the number of subjects treated within each group.

139 of 627 subjects (22.2 per cent) died. The primary cause recorded for each is below. Every death has a dated record in the disposition dataset and, where an adverse event was the cause, a fatal event in the adverse event dataset.

Disease progression. Recorded as the primary cause for 62 subjects (44.6 per cent of those who died). These deaths were not attributed to an adverse event and are recorded as disease outcomes.

Adverse event. Recorded as the primary cause for 68 subjects (48.9 per cent of those who died). Each of these subjects has a corresponding fatal adverse event and a narrative at Section 14.3.3.

Other. Recorded as the primary cause for 9 subjects (6.5 per cent of those who died). These deaths were not attributed to an adverse event and are recorded as disease outcomes.

12.3.2 Serious adverse events

Table 12-8 Serious Treatment-emergent Adverse Events by Preferred Term – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Anaemia	35 (8.4)	25 (12.0)	60 (9.6)
Pneumonia	11 (2.6)	4 (1.9)	15 (2.4)
Thrombocytopenia	13 (3.1)	1 (0.5)	14 (2.2)
Malignant neoplasm progression	6 (1.4)	7 (3.3)	13 (2.1)
Sepsis	7 (1.7)	4 (1.9)	11 (1.8)
Neutropenia	4 (1.0)	4 (1.9)	8 (1.3)
Pneumonitis	4 (1.0)	3 (1.4)	7 (1.1)
Myelodysplastic syndrome	3 (0.7)	3 (1.4)	6 (1.0)
Fatigue	3 (0.7)	1 (0.5)	4 (0.6)
Intestinal obstruction	2 (0.5)	2 (1.0)	4 (0.6)
Nausea	3 (0.7)	1 (0.5)	4 (0.6)
Ascites	2 (0.5)	1 (0.5)	3 (0.5)
Leukopenia	1 (0.2)	2 (1.0)	3 (0.5)
Alanine aminotransferase increased	1 (0.2)	1 (0.5)	2 (0.3)
Dyspnoea	2 (0.5)	0 (0.0)	2 (0.3)
Hypertension	2 (0.5)	0 (0.0)	2 (0.3)
Pulmonary embolism	1 (0.2)	1 (0.5)	2 (0.3)
Aspartate aminotransferase increased	1 (0.2)	0 (0.0)	1 (0.2)
Decreased appetite	0 (0.0)	1 (0.5)	1 (0.2)
Diarrhoea	1 (0.2)	0 (0.0)	1 (0.2)
Dizziness	0 (0.0)	1 (0.5)	1 (0.2)
Lymphopenia	1 (0.2)	0 (0.0)	1 (0.2)
Pleural effusion	0 (0.0)	1 (0.5)	1 (0.2)
Urinary tract infection	1 (0.2)	0 (0.0)	1 (0.2)
Vomiting	1 (0.2)	0 (0.0)	1 (0.2)

Subjects are counted once per preferred term. Percentages are based on N = 627.

12.3.3 Events leading to discontinuation

Table 12-9 Treatment-emergent Adverse Events Leading to Discontinuation of Study Treatment – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Anaemia	11 (2.6)	8 (3.8)	19 (3.0)
Malignant neoplasm progression	6 (1.4)	7 (3.3)	13 (2.1)
Sepsis	8 (1.9)	3 (1.4)	11 (1.8)
Neutropenia	5 (1.2)	3 (1.4)	8 (1.3)
Pneumonia	7 (1.7)	1 (0.5)	8 (1.3)
Thrombocytopenia	7 (1.7)	0 (0.0)	7 (1.1)
Intestinal obstruction	2 (0.5)	2 (1.0)	4 (0.6)
Ascites	2 (0.5)	1 (0.5)	3 (0.5)
Dyspnoea	3 (0.7)	0 (0.0)	3 (0.5)
Leukopenia	1 (0.2)	2 (1.0)	3 (0.5)
Myelodysplastic syndrome	1 (0.2)	2 (1.0)	3 (0.5)
Pneumonitis	1 (0.2)	2 (1.0)	3 (0.5)
Fatigue	2 (0.5)	0 (0.0)	2 (0.3)
Lymphopenia	2 (0.5)	0 (0.0)	2 (0.3)
Pulmonary embolism	1 (0.2)	1 (0.5)	2 (0.3)
Abdominal pain	1 (0.2)	0 (0.0)	1 (0.2)
Asthenia	1 (0.2)	0 (0.0)	1 (0.2)
Constipation	0 (0.0)	1 (0.5)	1 (0.2)
Decreased appetite	1 (0.2)	0 (0.0)	1 (0.2)
Dizziness	0 (0.0)	1 (0.5)	1 (0.2)
Nausea	1 (0.2)	0 (0.0)	1 (0.2)
Pleural effusion	0 (0.0)	1 (0.5)	1 (0.2)
Pyrexia	1 (0.2)	0 (0.0)	1 (0.2)

Percentages are based on N = 627.

12.3.4 Narratives

Narratives for every death, every serious adverse event and every withdrawal due to an adverse event are provided at Section 14.3.3, bound separately.

12.4 Clinical laboratory evaluation

Laboratory results were graded against the central laboratory reference ranges. Shifts from baseline to the worst post-baseline grade are in Table 12-10. Individual laboratory data are listed at Appendix 16.2.10.

Table 12-10 Shift from Baseline to Worst Post-baseline CTCAE Grade for Selected Laboratory Parameters – Safety Analysis Set

Laboratory parameter	Kelvaparib 300 mg: worst grade 3 n (%)	Kelvaparib 300 mg: worst grade 4 n (%)	Kelvaparib 300 mg: shift >= 2 grades n (%)	Placebo: worst grade 3 n (%)	Placebo: worst grade 4 n (%)	Placebo: shift >= 2 grades n (%)	Total: worst grade 3 n (%)	Total: worst grade 4 n (%)	Total: shift >= 2 grades n (%)
Haemoglobin	60 (14.4)	15 (3.6)	137 (32.8)	36 (17.2)	12 (5.7)	78 (37.3)	96 (15.3)	27 (4.3)	215 (34.3)
Platelets	30 (7.2)	7 (1.7)	80 (19.1)	21 (10.0)	2 (1.0)	36 (17.2)	51 (8.1)	9 (1.4)	116 (18.5)
Neutrophils	35 (8.4)	16 (3.8)	84 (20.1)	20 (9.6)	11 (5.3)	47 (22.5)	55 (8.8)	27 (4.3)	131 (20.9)
Lymphocytes	34 (8.1)	12 (2.9)	74 (17.7)	12 (5.7)	4 (1.9)	36 (17.2)	46 (7.3)	16 (2.6)	110 (17.5)
CA-125	55 (13.2)	12 (2.9)	136 (32.5)	35 (16.7)	10 (4.8)	67 (32.1)	90 (14.4)	22 (3.5)	203 (32.4)
ALT	23 (5.5)	3 (0.7)	50 (12.0)	11 (5.3)	1 (0.5)	19 (9.1)	34 (5.4)	4 (0.6)	69 (11.0)
AST	14 (3.3)	1 (0.2)	33 (7.9)	9 (4.3)	3 (1.4)	19 (9.1)	23 (3.7)	4 (0.6)	52 (8.3)
Creatinine	22 (5.3)	8 (1.9)	49 (11.7)	10 (4.8)	1 (0.5)	19 (9.1)	32 (5.1)	9 (1.4)	68 (10.8)
Lactate dehydrogenase	26 (6.2)	5 (1.2)	58 (13.9)	14 (6.7)	4 (1.9)	24 (11.5)	40 (6.4)	9 (1.4)	82 (13.1)
Albumin	14 (3.3)	4 (1.0)	27 (6.5)	5 (2.4)	6 (2.9)	23 (11.0)	19 (3.0)	10 (1.6)	50 (8.0)

CTCAE = Common Terminology Criteria for Adverse Events, version 5.0. Percentages are based on the number of subjects treated in each group. Overall N = 627.

12.4.1 Findings for each laboratory test

ALT. An abnormality of any grade was present at baseline in 183 subjects (29.2 per cent) and post-baseline in 281 (44.8 per cent). 135 subjects (21.5 per cent) shifted to a worse grade than their own baseline, and 38 (6.1 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

AST. An abnormality of any grade was present at baseline in 155 subjects (24.7 per cent) and post-baseline in 251 (40.0 per cent). 117 subjects (18.7 per cent) shifted to a worse grade than their own baseline, and 27 (4.3 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Albumin. An abnormality of any grade was present at baseline in 167 subjects (26.6 per cent) and post-baseline in 243 (38.8 per cent). 111 subjects (17.7 per cent) shifted to a worse grade than their own baseline, and 29 (4.6 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

CA-125. An abnormality of any grade was present at baseline in 169 subjects (27.0 per cent) and post-baseline in 485 (77.4 per cent). 438 subjects (69.9 per cent) shifted to a worse grade than their own baseline, and 112 (17.9 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Creatinine. An abnormality of any grade was present at baseline in 202 subjects (32.2 per cent) and post-baseline in 304 (48.5 per cent). 147 subjects (23.4 per cent) shifted to a worse grade than their own baseline, and 41 (6.5 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Haemoglobin. An abnormality of any grade was present at baseline in 193 subjects (30.8 per cent) and post-baseline in 514 (82.0 per cent). 454 subjects (72.4 per cent) shifted to a worse grade than their own baseline, and 123 (19.6 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Lactate dehydrogenase. An abnormality of any grade was present at baseline in 186 subjects (29.7 per cent) and post-baseline in 316 (50.4 per cent). 184 subjects (29.3 per cent) shifted to a worse grade than their own baseline, and 49 (7.8 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Lymphocytes. An abnormality of any grade was present at baseline in 189 subjects (30.1 per cent) and post-baseline in 350 (55.8 per cent). 229 subjects (36.5 per cent) shifted to a worse grade than their own baseline, and 62 (9.9 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Neutrophils. An abnormality of any grade was present at baseline in 197 subjects (31.4 per cent) and post-baseline in 397 (63.3 per cent). 286 subjects (45.6 per cent) shifted to a worse grade than their own baseline, and 82 (13.1 per cent) reached grade 3 or higher at

some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Platelets. An abnormality of any grade was present at baseline in 164 subjects (26.2 per cent) and post-baseline in 370 (59.0 per cent). 270 subjects (43.1 per cent) shifted to a worse grade than their own baseline, and 60 (9.6 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

12.5 Vital signs, physical findings and other safety observations

No clinically significant pattern of change in vital signs was observed. Individual vital sign data are listed at Appendix 16.2.11.

12.5.1 Electrocardiograms

The protocol specified electrocardiograms, but no electrocardiogram data are held in the study database, so no electrocardiogram results are reported.

12.6 Safety conclusions

The safety profile observed was consistent with that anticipated for oral PARP1-selective inhibitor in this population. The events that occurred most often, and those that most often reached grade 3 or higher, are those for which the protocol required specific monitoring and management. No new safety concern was identified.

13 DISCUSSION AND OVERALL CONCLUSIONS

Study KLV-7701 evaluated kelvaparib in recurrent ovarian carcinoma after prior platinum-based therapy, platinum-sensitive or platinum-resistant. Kelvaparib prolonged progression-free survival compared with placebo in recurrent ovarian carcinoma after prior platinum-based therapy, platinum-sensitive or platinum-resistant (stratified hazard ratio 0.64, 95 per cent CI 0.50 to 0.81; stratified log-rank p = 0.0002), and the primary endpoint was met. The objective response rate (54.5 per cent) was also significantly higher with kelvaparib (Cochran-Mantel-Haenszel p < 0.0001); overall survival, the next endpoint in the testing hierarchy, did not differ (stratified log-rank p = 0.8651), and findings below that point are descriptive. The safety profile was dominated by anaemia and myelosuppression, both on-target consequences of the mechanism of action. Myelodysplastic syndrome or acute myeloid leukaemia occurred in 1.4 per cent of subjects (Section 12.2.6). Limitations should be borne in mind. The study was open-label, which may influence investigator assessment of response and reporting of subjective adverse events. Follow-up was ongoing at the data cut-off and time-to-event estimates are therefore immature. Taken together, the results of study KLV-7701 support the continued development of kelvaparib at 300 mg, with weekly blood counts for the first month and monthly thereafter, in high-grade serous ovarian carcinoma.

The population enrolled was consistent with the intended population and with the eligibility criteria. Exposure was sufficient to characterise both the efficacy and the safety of the regimen. The primary analysis was performed as planned; the sensitivity analyses the statistical analysis plan lists are not reported here.

The population enrolled was the population the protocol described. Median age, performance status and prior therapy were consistent with the eligibility criteria, and there is no indication that the study enrolled a population easier to treat than the one intended.

Exposure was adequate. The median relative dose intensity indicates that the regimen as written could be delivered, and the pattern of reduction and interruption was consistent with the toxicities anticipated for oral PARP1-selective inhibitor.

The efficacy result should be read together with its confidence interval and with the maturity of the follow-up. Because the study was randomised, the comparison is internally controlled and the estimate does not depend on an external assumption about how this population would otherwise have fared.

The safety profile was consistent with the mechanism and with the earlier clinical experience. The events reported most often, and those most often reaching grade 3 or higher, were those for which the protocol required specific monitoring, which indicates that the monitoring was directed at the right things.

96 subjects (15.3 per cent) discontinued study treatment because of an adverse event. That figure, rather than the overall incidence of events, is the measure of whether the regimen is tolerable in practice, because it counts the events that could not be managed by dose modification.

Deaths were reviewed individually. Those attributed to an adverse event are described in the narratives; the remainder are recorded as disease outcomes and are consistent with what would be expected in this population over this period.

The limitations of this study should be borne in mind. Follow-up at this analysis is limited and the longer time-to-event estimates are immature. The study was open-label, so investigator assessment of any subjective endpoint may have been influenced by knowledge of the assigned treatment. The subgroup analyses are descriptive.

Taken together, the results support the conclusions stated above and do not identify a safety concern that was not anticipated. Nothing in this analysis indicates that the assumptions underlying the design were materially wrong.

The limitations of the study should be borne in mind when interpreting these results. The study was open-label, so investigator assessment of subjective endpoints may have been influenced by knowledge of the assigned treatment. Follow-up at the time of this analysis is limited, so the estimates of the longer time-to-event endpoints are immature and will change as further events accrue.

14 TABLES AND FIGURES REFERRED TO BUT NOT IN THE TEXT

The summary tables referred to in the text are presented in this section. Subject data listings are provided in Appendix 16.2.

14.1 Demographic data

Table 14.1.1 Demographic and Baseline Characteristics – Safety Analysis Set

Characteristic	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Age (years)			
Mean (SD)	65.3 (9.00)	64.7 (9.21)	65.1 (9.07)
Median	65.0	65.0	65.0
Min, max	38, 88	42, 88	38, 88
>= 65 years, n (%)	225 (53.8)	108 (51.7)	333 (53.1)
Sex, n (%)			
Male	0 (0.0)	0 (0.0)	0 (0.0)
Female	418 (100.0)	209 (100.0)	627 (100.0)
Race, n (%)			
White	342 (81.8)	161 (77.0)	503 (80.2)
Asian	38 (9.1)	25 (12.0)	63 (10.0)
Black or African American	20 (4.8)	11 (5.3)	31 (4.9)
Other	18 (4.3)	12 (5.7)	30 (4.8)
ECOG performance status, n (%)			
0	193 (46.2)	103 (49.3)	296 (47.2)
1	184 (44.0)	89 (42.6)	273 (43.5)
2	41 (9.8)	17 (8.1)	58 (9.3)
BRCA mutation status, n (%)			
BRCA1 mutated	101 (24.2)	49 (23.4)	150 (23.9)
BRCA2 mutated	62 (14.8)	33 (15.8)	95 (15.2)
BRCA wild-type	255 (61.0)	127 (60.8)	382 (60.9)
Homologous recombination deficiency, n (%)	222 (53.1)	104 (49.8)	326 (52.0)
Platinum-sensitive disease, n (%)	265 (63.4)	135 (64.6)	400 (63.8)
Residual disease after cytoreductive surgery, n (%)	178 (42.6)	88 (42.1)	266 (42.4)
Number of prior lines of platinum-based therapy			
Median	2.0	2.0	2.0

Characteristic	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
>= 3, n (%)	122 (29.2)	61 (29.2)	183 (29.2)

SD = standard deviation; ECOG = Eastern Cooperative Oncology Group. Percentages are based on the number of subjects in the safety analysis set within each group.

Table 14.1.2 Subject Disposition – Safety Analysis Set

Disposition	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Subjects treated, n	418	209	627
Treatment ongoing at data cut-off, n (%)	101 (24.2)	43 (20.6)	144 (23.0)
Discontinued study treatment, n (%)	317 (75.8)	166 (79.4)	483 (77.0)
Disease progression	37 (8.9)	41 (19.6)	78 (12.4)
Adverse event	25 (6.0)	12 (5.7)	37 (5.9)
Death	38 (9.1)	23 (11.0)	61 (9.7)
Withdrawal by subject	82 (19.6)	37 (17.7)	119 (19.0)
Physician decision	67 (16.0)	27 (12.9)	94 (15.0)
Lost to follow-up	39 (9.3)	12 (5.7)	51 (8.1)
Protocol deviation	29 (6.9)	14 (6.7)	43 (6.9)

Percentages are based on the number of subjects treated within each group.

Table 14.1.3 Extent of Exposure – Safety Analysis Set

Exposure parameter	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Duration of treatment (days)			
Median	122.0	116.0	119.0
Q1, Q3	70.2, 166.5	62.0, 172.0	68.0, 169.0
Min, max	8, 340	8, 353	8, 353
Cycles started, median	4.0	4.0	4.0
Relative dose intensity (%), median	86.8	87.8	86.8
Any dose reduction, n (%)	73 (17.5)	40 (19.1)	113 (18.0)
Any dose interruption, n (%)	122 (29.2)	59 (28.2)	181 (28.9)

Q1 = first quartile; Q3 = third quartile. Relative dose intensity is the total dose received divided by the total dose planned over the treatment period.

14.2 Efficacy data

Table 14.2.1 Best Overall Response by RECIST 1.1 – Full Analysis Set

Best overall response	Kelvaparib 300 mg (N=405)	Placebo (N=199)	Total (N=604)
Complete response (CR)	49 (12.1)	15 (7.5)	64 (10.6)
Partial response (PR)	201 (49.6)	64 (32.2)	265 (43.9)
Stable disease	112 (27.7)	76 (38.2)	188 (31.1)
Progressive disease	43 (10.6)	44 (22.1)	87 (14.4)
Not evaluable	0 (0.0)	0 (0.0)	0 (0.0)
Objective response rate, n (%)	250 (61.7)	79 (39.7)	329 (54.5)
95% CI	(56.8, 66.5)	(32.8, 46.9)	(50.4, 58.5)

CI = confidence interval. Assessed according to RECIST version 1.1. Confidence intervals are exact (Clopper-Pearson).

Table 14.2.2 Kaplan-Meier Estimates of Time-to-event Endpoints – Full Analysis Set

Endpoint	Kelvaparib 300 mg (N=405)	Placebo (N=199)	Total (N=604)
Duration of response, median (months)	12.0	9.9	11.7
25th, 75th percentile	(5.8, 16.2)	(3.3, 13.7)	(5.3, 15.9)
Progression-free survival, median (months)	9.2	6.7	8.2
25th, 75th percentile	(3.6, 14.8)	(2.6, 11.3)	(3.1, 13.2)
Overall survival, median (months)	28.5	30.4	30.4
25th, 75th percentile	(19.5, NE)	(24.5, NE)	(19.6, NE)

NE = not estimable. Duration of response is analysed among responding subjects only. It is measured from the first dose to the end of response recorded in the analysis dataset; response and progression dates cannot be re-derived because no assessment-level tumour data were collected in the database.

Table 14.2.3 Overall Response Rate and Progression-free Survival by BRCA Mutation Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
BRCA wild-type	367	197 (53.7)	(48.4, 58.9)	8.2
BRCA1 mutated	146	82 (56.2)	(47.7, 64.4)	8.0
BRCA2 mutated	91	50 (54.9)	(44.2, 65.4)	8.4

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.4 Overall Response Rate and Progression-free Survival by Homologous Recombination Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
Negative	291	148 (50.9)	(45.0, 56.7)	7.5
Positive	313	181 (57.8)	(52.1, 63.4)	8.9

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.5 Overall Response Rate and Progression-free Survival by Platinum Sensitivity – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
No	220	127 (57.7)	(50.9, 64.3)	8.0
Yes	384	202 (52.6)	(47.5, 57.7)	8.3

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.6 Overall Response Rate and Progression-free Survival by Residual Disease After Surgery – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
Absent	351	190 (54.1)	(48.8, 59.4)	8.5
Present	253	139 (54.9)	(48.6, 61.2)	7.4

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.7 Overall Response Rate and Progression-free Survival by Prior Lines of Platinum-based Therapy – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
1 to 2	426	229 (53.8)	(48.9, 58.6)	8.5
>= 3	178	100 (56.2)	(48.6, 63.6)	7.6

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.8 Overall Response Rate and Progression-free Survival by Age Category – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
< 65 years	281	165 (58.7)	(52.7, 64.5)	8.1
>= 65 years	323	164 (50.8)	(45.2, 56.4)	8.3

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.9 Overall Response Rate and Progression-free Survival by ECOG Performance Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
0	283	160 (56.5)	(50.5, 62.4)	7.6
1	265	138 (52.1)	(45.9, 58.2)	9.0
2	56	31 (55.4)	(41.5, 68.7)	8.1

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

14.3 Safety data

Table 14.3.1 Overview of Treatment-emergent Adverse Events – Safety Analysis Set

Category, n (%)	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Any TEAE	417 (99.8)	209 (100.0)	626 (99.8)
Any treatment-related TEAE	411 (98.3)	204 (97.6)	615 (98.1)
Any grade >= 3 TEAE	190 (45.5)	108 (51.7)	298 (47.5)
Any serious TEAE	98 (23.4)	61 (29.2)	159 (25.4)
Any TEAE leading to dose reduction	72 (17.2)	39 (18.7)	111 (17.7)
Any TEAE leading to discontinuation	61 (14.6)	35 (16.7)	96 (15.3)
Any fatal TEAE	43 (10.3)	25 (12.0)	68 (10.8)

TEAE = treatment-emergent adverse event. A subject is counted once within each category.

Table 14.3.2 Treatment-emergent Adverse Events Occurring in at Least 2% of Subjects by System Organ Class and Preferred Term – Safety Analysis Set

System organ class / Preferred term	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Gastrointestinal disorders	364 (87.1)	180 (86.1)	544 (86.8)
Nausea	257 (61.5)	136 (65.1)	393 (62.7)
Vomiting	113 (27.0)	50 (23.9)	163 (26.0)
Constipation	105 (25.1)	58 (27.8)	163 (26.0)
Diarrhoea	81 (19.4)	38 (18.2)	119 (19.0)
Abdominal pain	84 (20.1)	34 (16.3)	118 (18.8)
Dyspepsia	44 (10.5)	23 (11.0)	67 (10.7)
Blood and lymphatic system disorders	310 (74.2)	162 (77.5)	472 (75.3)
Anaemia	202 (48.3)	112 (53.6)	314 (50.1)
Thrombocytopenia	101 (24.2)	50 (23.9)	151 (24.1)
Neutropenia	83 (19.9)	43 (20.6)	126 (20.1)
Leukopenia	44 (10.5)	29 (13.9)	73 (11.6)
Lymphopenia	25 (6.0)	20 (9.6)	45 (7.2)
General disorders and administration site conditions	253 (60.5)	129 (61.7)	382 (60.9)
Fatigue	198 (47.4)	101 (48.3)	299 (47.7)
Asthenia	54 (12.9)	33 (15.8)	87 (13.9)
Oedema peripheral	33 (7.9)	17 (8.1)	50 (8.0)
Pyrexia	20 (4.8)	10 (4.8)	30 (4.8)
Nervous system disorders	185 (44.3)	94 (45.0)	279 (44.5)
Headache	85 (20.3)	42 (20.1)	127 (20.3)
Dysgeusia	67 (16.0)	48 (23.0)	115 (18.3)
Dizziness	69 (16.5)	29 (13.9)	98 (15.6)
Musculoskeletal and connective tissue disorders	123 (29.4)	68 (32.5)	191 (30.5)
Back pain	50 (12.0)	30 (14.4)	80 (12.8)
Arthralgia	48 (11.5)	26 (12.4)	74 (11.8)
Myalgia	33 (7.9)	18 (8.6)	51 (8.1)
Infections and infestations	118 (28.2)	62 (29.7)	180 (28.7)

System organ class / Preferred term	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Nasopharyngitis	55 (13.2)	22 (10.5)	77 (12.3)
Urinary tract infection	39 (9.3)	25 (12.0)	64 (10.2)
Pneumonia	24 (5.7)	14 (6.7)	38 (6.1)
Sepsis	12 (2.9)	10 (4.8)	22 (3.5)
Investigations	106 (25.4)	57 (27.3)	163 (26.0)
Alanine aminotransferase increased	37 (8.9)	20 (9.6)	57 (9.1)
Aspartate aminotransferase increased	28 (6.7)	16 (7.7)	44 (7.0)
Weight decreased	28 (6.7)	13 (6.2)	41 (6.5)
Blood creatinine increased	26 (6.2)	14 (6.7)	40 (6.4)
Metabolism and nutrition disorders	95 (22.7)	57 (27.3)	152 (24.2)
Decreased appetite	95 (22.7)	57 (27.3)	152 (24.2)
Respiratory, thoracic and mediastinal disorders	96 (23.0)	47 (22.5)	143 (22.8)
Cough	55 (13.2)	26 (12.4)	81 (12.9)
Dyspnoea	34 (8.1)	16 (7.7)	50 (8.0)
Pneumonitis	14 (3.3)	7 (3.3)	21 (3.3)
Psychiatric disorders	46 (11.0)	30 (14.4)	76 (12.1)
Insomnia	46 (11.0)	30 (14.4)	76 (12.1)
Skin and subcutaneous tissue disorders	30 (7.2)	27 (12.9)	57 (9.1)
Rash	30 (7.2)	27 (12.9)	57 (9.1)
Vascular disorders	27 (6.5)	10 (4.8)	37 (5.9)
Hypertension	27 (6.5)	10 (4.8)	37 (5.9)
Neoplasms benign, malignant and unspecified	11 (2.6)	11 (5.3)	22 (3.5)
Malignant neoplasm progression	6 (1.4)	7 (3.3)	13 (2.1)

Subjects are counted once per preferred term and once per system organ class at their highest reported grade. MedDRA version 25.0.

Table 14.3.3 Treatment-emergent Adverse Events of Grade 3 or Higher by System Organ Class and Preferred Term – Safety Analysis Set

System organ class / Preferred term	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Blood and lymphatic system disorders	110 (26.3)	63 (30.1)	173 (27.6)
Anaemia	60 (14.4)	43 (20.6)	103 (16.4)
Neutropenia	22 (5.3)	11 (5.3)	33 (5.3)
Thrombocytopenia	22 (5.3)	11 (5.3)	33 (5.3)
Lymphopenia	6 (1.4)	3 (1.4)	9 (1.4)
Leukopenia	5 (1.2)	4 (1.9)	9 (1.4)
Infections and infestations	26 (6.2)	15 (7.2)	41 (6.5)
Pneumonia	13 (3.1)	5 (2.4)	18 (2.9)
Sepsis	10 (2.4)	8 (3.8)	18 (2.9)
Urinary tract infection	2 (0.5)	2 (1.0)	4 (0.6)
Nasopharyngitis	1 (0.2)	0 (0.0)	1 (0.2)
Gastrointestinal disorders	27 (6.5)	14 (6.7)	41 (6.5)
Nausea	12 (2.9)	7 (3.3)	19 (3.0)
Vomiting	6 (1.4)	2 (1.0)	8 (1.3)
Intestinal obstruction	2 (0.5)	2 (1.0)	4 (0.6)
Ascites	2 (0.5)	1 (0.5)	3 (0.5)
Diarrhoea	3 (0.7)	0 (0.0)	3 (0.5)
Abdominal pain	2 (0.5)	1 (0.5)	3 (0.5)
Constipation	1 (0.2)	1 (0.5)	2 (0.3)
General disorders and administration site conditions	15 (3.6)	8 (3.8)	23 (3.7)
Fatigue	8 (1.9)	7 (3.3)	15 (2.4)
Asthenia	5 (1.2)	0 (0.0)	5 (0.8)
Pyrexia	2 (0.5)	0 (0.0)	2 (0.3)
Oedema peripheral	0 (0.0)	1 (0.5)	1 (0.2)
Neoplasms benign, malignant and unspecified	9 (2.2)	11 (5.3)	20 (3.2)
Malignant neoplasm progression	6 (1.4)	7 (3.3)	13 (2.1)
Myelodysplastic syndrome	3 (0.7)	4 (1.9)	7 (1.1)

System organ class / Preferred term	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Respiratory, thoracic and mediastinal disorders	9 (2.2)	8 (3.8)	17 (2.7)
Pneumonitis	4 (1.0)	3 (1.4)	7 (1.1)
Dyspnoea	4 (1.0)	2 (1.0)	6 (1.0)
Pulmonary embolism	1 (0.2)	1 (0.5)	2 (0.3)
Pleural effusion	0 (0.0)	1 (0.5)	1 (0.2)
Cough	0 (0.0)	1 (0.5)	1 (0.2)
Investigations	8 (1.9)	5 (2.4)	13 (2.1)
Alanine aminotransferase increased	2 (0.5)	5 (2.4)	7 (1.1)
Blood creatinine increased	5 (1.2)	0 (0.0)	5 (0.8)
Aspartate aminotransferase increased	2 (0.5)	0 (0.0)	2 (0.3)
Vascular disorders	4 (1.0)	1 (0.5)	5 (0.8)
Hypertension	4 (1.0)	1 (0.5)	5 (0.8)
Nervous system disorders	2 (0.5)	3 (1.4)	5 (0.8)
Headache	2 (0.5)	1 (0.5)	3 (0.5)
Dizziness	0 (0.0)	2 (1.0)	2 (0.3)
Metabolism and nutrition disorders	1 (0.2)	2 (1.0)	3 (0.5)
Decreased appetite	1 (0.2)	2 (1.0)	3 (0.5)
Skin and subcutaneous tissue disorders	2 (0.5)	0 (0.0)	2 (0.3)
Rash	2 (0.5)	0 (0.0)	2 (0.3)
Musculoskeletal and connective tissue disorders	1 (0.2)	0 (0.0)	1 (0.2)
Arthralgia	1 (0.2)	0 (0.0)	1 (0.2)
Psychiatric disorders	1 (0.2)	0 (0.0)	1 (0.2)
Insomnia	1 (0.2)	0 (0.0)	1 (0.2)

Subjects are counted once per preferred term and once per system organ class at their highest reported grade. MedDRA version 25.0.

Table 14.3.4 Treatment-related Treatment-emergent Adverse Events Occurring in at Least 2% of Subjects – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Nausea	179 (42.8)	84 (40.2)	263 (41.9)
Anaemia	117 (28.0)	75 (35.9)	192 (30.6)
Fatigue	106 (25.4)	50 (23.9)	156 (24.9)
Vomiting	73 (17.5)	34 (16.3)	107 (17.1)
Constipation	69 (16.5)	34 (16.3)	103 (16.4)
Decreased appetite	64 (15.3)	34 (16.3)	98 (15.6)
Thrombocytopenia	57 (13.6)	30 (14.4)	87 (13.9)
Headache	56 (13.4)	28 (13.4)	84 (13.4)
Neutropenia	50 (12.0)	30 (14.4)	80 (12.8)
Diarrhoea	51 (12.2)	29 (13.9)	80 (12.8)
Abdominal pain	48 (11.5)	25 (12.0)	73 (11.6)
Dysgeusia	43 (10.3)	27 (12.9)	70 (11.2)
Dizziness	40 (9.6)	21 (10.0)	61 (9.7)
Asthenia	37 (8.9)	19 (9.1)	56 (8.9)
Nasopharyngitis	37 (8.9)	15 (7.2)	52 (8.3)
Cough	35 (8.4)	17 (8.1)	52 (8.3)
Back pain	31 (7.4)	19 (9.1)	50 (8.0)
Insomnia	29 (6.9)	17 (8.1)	46 (7.3)
Arthralgia	24 (5.7)	20 (9.6)	44 (7.0)
Leukopenia	25 (6.0)	18 (8.6)	43 (6.9)
Urinary tract infection	28 (6.7)	14 (6.7)	42 (6.7)
Dyspepsia	26 (6.2)	15 (7.2)	41 (6.5)
Rash	18 (4.3)	21 (10.0)	39 (6.2)
Alanine aminotransferase increased	21 (5.0)	13 (6.2)	34 (5.4)
Dyspnoea	24 (5.7)	6 (2.9)	30 (4.8)
Myalgia	20 (4.8)	10 (4.8)	30 (4.8)
Oedema peripheral	22 (5.3)	7 (3.3)	29 (4.6)

Preferred term	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Blood creatinine increased	19 (4.5)	10 (4.8)	29 (4.6)
Hypertension	20 (4.8)	8 (3.8)	28 (4.5)
Weight decreased	20 (4.8)	7 (3.3)	27 (4.3)
Aspartate aminotransferase increased	15 (3.6)	10 (4.8)	25 (4.0)
Lymphopenia	11 (2.6)	13 (6.2)	24 (3.8)
Pneumonia	15 (3.6)	9 (4.3)	24 (3.8)
Sepsis	9 (2.2)	10 (4.8)	19 (3.0)
Pyrexia	11 (2.6)	5 (2.4)	16 (2.6)

Relationship assessed by the investigator. Percentages are based on the number of subjects treated in each group. Overall N = 627.

Table 14.3.5 Serious Treatment-emergent Adverse Events by Preferred Term – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Anaemia	35 (8.4)	25 (12.0)	60 (9.6)
Pneumonia	11 (2.6)	4 (1.9)	15 (2.4)
Thrombocytopenia	13 (3.1)	1 (0.5)	14 (2.2)
Malignant neoplasm progression	6 (1.4)	7 (3.3)	13 (2.1)
Sepsis	7 (1.7)	4 (1.9)	11 (1.8)
Neutropenia	4 (1.0)	4 (1.9)	8 (1.3)
Pneumonitis	4 (1.0)	3 (1.4)	7 (1.1)
Myelodysplastic syndrome	3 (0.7)	3 (1.4)	6 (1.0)
Fatigue	3 (0.7)	1 (0.5)	4 (0.6)
Intestinal obstruction	2 (0.5)	2 (1.0)	4 (0.6)
Nausea	3 (0.7)	1 (0.5)	4 (0.6)
Ascites	2 (0.5)	1 (0.5)	3 (0.5)
Leukopenia	1 (0.2)	2 (1.0)	3 (0.5)
Alanine aminotransferase increased	1 (0.2)	1 (0.5)	2 (0.3)
Dyspnoea	2 (0.5)	0 (0.0)	2 (0.3)
Hypertension	2 (0.5)	0 (0.0)	2 (0.3)
Pulmonary embolism	1 (0.2)	1 (0.5)	2 (0.3)
Aspartate aminotransferase increased	1 (0.2)	0 (0.0)	1 (0.2)
Decreased appetite	0 (0.0)	1 (0.5)	1 (0.2)
Diarrhoea	1 (0.2)	0 (0.0)	1 (0.2)
Dizziness	0 (0.0)	1 (0.5)	1 (0.2)
Lymphopenia	1 (0.2)	0 (0.0)	1 (0.2)
Pleural effusion	0 (0.0)	1 (0.5)	1 (0.2)
Urinary tract infection	1 (0.2)	0 (0.0)	1 (0.2)
Vomiting	1 (0.2)	0 (0.0)	1 (0.2)

Subjects are counted once per preferred term. Percentages are based on N = 627.

Table 14.3.6 Treatment-emergent Adverse Events Leading to Discontinuation of Study Treatment – Safety Analysis Set

Preferred term	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Anaemia	11 (2.6)	8 (3.8)	19 (3.0)
Malignant neoplasm progression	6 (1.4)	7 (3.3)	13 (2.1)
Sepsis	8 (1.9)	3 (1.4)	11 (1.8)
Neutropenia	5 (1.2)	3 (1.4)	8 (1.3)
Pneumonia	7 (1.7)	1 (0.5)	8 (1.3)
Thrombocytopenia	7 (1.7)	0 (0.0)	7 (1.1)
Intestinal obstruction	2 (0.5)	2 (1.0)	4 (0.6)
Ascites	2 (0.5)	1 (0.5)	3 (0.5)
Dyspnoea	3 (0.7)	0 (0.0)	3 (0.5)
Leukopenia	1 (0.2)	2 (1.0)	3 (0.5)
Myelodysplastic syndrome	1 (0.2)	2 (1.0)	3 (0.5)
Pneumonitis	1 (0.2)	2 (1.0)	3 (0.5)
Fatigue	2 (0.5)	0 (0.0)	2 (0.3)
Lymphopenia	2 (0.5)	0 (0.0)	2 (0.3)
Pulmonary embolism	1 (0.2)	1 (0.5)	2 (0.3)
Abdominal pain	1 (0.2)	0 (0.0)	1 (0.2)
Asthenia	1 (0.2)	0 (0.0)	1 (0.2)
Constipation	0 (0.0)	1 (0.5)	1 (0.2)
Decreased appetite	1 (0.2)	0 (0.0)	1 (0.2)
Dizziness	0 (0.0)	1 (0.5)	1 (0.2)
Nausea	1 (0.2)	0 (0.0)	1 (0.2)
Pleural effusion	0 (0.0)	1 (0.5)	1 (0.2)
Pyrexia	1 (0.2)	0 (0.0)	1 (0.2)

Percentages are based on N = 627.

Table 14.3.7 Deaths – Safety Analysis Set

Category, n (%)	Kelvaparib 300 mg (N=418)	Placebo (N=209)	Total (N=627)
Any death	95 (22.7)	44 (21.1)	139 (22.2)
Death within 30 days of last dose	38 (9.1)	23 (11.0)	61 (9.7)
Primary cause of death			
Disease progression	47 (11.2)	15 (7.2)	62 (9.9)
Adverse event	43 (10.3)	25 (12.0)	68 (10.8)
Other	5 (1.2)	4 (1.9)	9 (1.4)

Percentages are based on the number of subjects treated within each group.

Table 14.3.8 Shift from Baseline to Worst Post-baseline CTCAE Grade for Selected Laboratory Parameters – Safety Analysis Set

Laboratory parameter	Kelvaparib 300 mg: worst grade 3 n (%)	Kelvaparib 300 mg: worst grade 4 n (%)	Kelvaparib 300 mg: shift >= 2 grades n (%)	Placebo: worst grade 3 n (%)	Placebo: worst grade 4 n (%)	Placebo: shift >= 2 grades n (%)	Total: worst grade 3 n (%)	Total: worst grade 4 n (%)	Total: shift >= 2 grades n (%)
Haemoglobin	60 (14.4)	15 (3.6)	137 (32.8)	36 (17.2)	12 (5.7)	78 (37.3)	96 (15.3)	27 (4.3)	215 (34.3)
Platelets	30 (7.2)	7 (1.7)	80 (19.1)	21 (10.0)	2 (1.0)	36 (17.2)	51 (8.1)	9 (1.4)	116 (18.5)
Neutrophils	35 (8.4)	16 (3.8)	84 (20.1)	20 (9.6)	11 (5.3)	47 (22.5)	55 (8.8)	27 (4.3)	131 (20.9)
Lymphocytes	34 (8.1)	12 (2.9)	74 (17.7)	12 (5.7)	4 (1.9)	36 (17.2)	46 (7.3)	16 (2.6)	110 (17.5)
CA-125	55 (13.2)	12 (2.9)	136 (32.5)	35 (16.7)	10 (4.8)	67 (32.1)	90 (14.4)	22 (3.5)	203 (32.4)
ALT	23 (5.5)	3 (0.7)	50 (12.0)	11 (5.3)	1 (0.5)	19 (9.1)	34 (5.4)	4 (0.6)	69 (11.0)
AST	14 (3.3)	1 (0.2)	33 (7.9)	9 (4.3)	3 (1.4)	19 (9.1)	23 (3.7)	4 (0.6)	52 (8.3)
Creatinine	22 (5.3)	8 (1.9)	49 (11.7)	10 (4.8)	1 (0.5)	19 (9.1)	32 (5.1)	9 (1.4)	68 (10.8)
Lactate dehydrogenase	26 (6.2)	5 (1.2)	58 (13.9)	14 (6.7)	4 (1.9)	24 (11.5)	40 (6.4)	9 (1.4)	82 (13.1)
Albumin	14 (3.3)	4 (1.0)	27 (6.5)	5 (2.4)	6 (2.9)	23 (11.0)	19 (3.0)	10 (1.6)	50 (8.0)

CTCAE = Common Terminology Criteria for Adverse Events, version 5.0. Percentages are based on the number of subjects treated in each group. Overall N = 627.

Table 14.3.9 Time to Onset and Resolution of the Most Frequent Treatment-emergent Adverse Events – Safety Analysis Set

Preferred term	Events, n	Median time to onset (days)	Resolved, n (%)	Median time to resolution (days)
Nausea	393	32.0	330 (84.0)	11.0
Anaemia	317	29.0	265 (83.6)	10.0
Fatigue	299	32.0	239 (79.9)	10.0
Vomiting	163	34.0	130 (79.8)	11.0
Constipation	163	29.0	131 (80.4)	9.0
Decreased appetite	152	38.0	127 (83.6)	10.0
Thrombocytopenia	152	41.0	129 (84.9)	12.0
Neutropenia	127	36.0	98 (77.2)	9.5
Headache	127	25.0	105 (82.7)	11.0
Diarrhoea	119	37.0	97 (81.5)	11.0
Abdominal pain	118	25.5	102 (86.4)	11.5
Dysgeusia	115	24.0	92 (80.0)	10.0
Dizziness	98	23.5	76 (77.6)	12.0
Asthenia	87	33.0	74 (85.1)	10.0
Cough	81	33.0	65 (80.2)	10.0
Back pain	80	28.0	67 (83.8)	10.0
Nasopharyngitis	77	29.0	61 (79.2)	11.0
Insomnia	76	38.0	65 (85.5)	9.0
Arthralgia	74	36.0	65 (87.8)	8.0
Leukopenia	73	30.0	57 (78.1)	11.0

Events are counted at the event level rather than the subject level. NE = not estimable.

14.3.10 Adverse events within subgroups

The summaries in Tables 14.3.10 to 14.3.16 are exploratory. The subgroups were not chosen or sized for comparisons of safety, the tables are not adjusted for multiplicity, and no conclusion about a difference between subgroups is drawn from them.

Table 14.3.10 Overview of adverse events by BRCA mutation status

Category	BRCA wild-type	BRCA1 mutated	BRCA2 mutated
Any event	382 (100.0)	149 (99.3)	95 (100.0)
Grade 3 or higher	188 (49.2)	70 (46.7)	40 (42.1)
Serious	102 (26.7)	34 (22.7)	23 (24.2)
Led to discontinuation	57 (14.9)	26 (17.3)	13 (13.7)
Fatal	40 (10.5)	20 (13.3)	8 (8.4)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.11 Overview of adverse events by homologous recombination status

Category	Negative	Positive
Any event	301 (100.0)	325 (99.7)
Grade 3 or higher	148 (49.2)	150 (46.0)
Serious	84 (27.9)	75 (23.0)
Led to discontinuation	46 (15.3)	50 (15.3)
Fatal	34 (11.3)	34 (10.4)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.12 Overview of adverse events by platinum sensitivity

Category	No	Yes
Any event	227 (100.0)	399 (99.8)
Grade 3 or higher	103 (45.4)	195 (48.8)
Serious	62 (27.3)	97 (24.2)
Led to discontinuation	35 (15.4)	61 (15.2)
Fatal	24 (10.6)	44 (11.0)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.13 Overview of adverse events by residual disease after surgery

Category	Absent	Present
Any event	361 (100.0)	265 (99.6)
Grade 3 or higher	166 (46.0)	132 (49.6)
Serious	83 (23.0)	76 (28.6)
Led to discontinuation	47 (13.0)	49 (18.4)
Fatal	32 (8.9)	36 (13.5)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.14 Overview of adverse events by prior lines of platinum-based therapy

Category	1 to 2	>= 3
Any event	443 (99.8)	183 (100.0)
Grade 3 or higher	196 (44.1)	102 (55.7)
Serious	95 (21.4)	64 (35.0)
Led to discontinuation	58 (13.1)	38 (20.8)
Fatal	41 (9.2)	27 (14.8)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.15 Overview of adverse events by age category

Category	<65	>=65
Any event	294 (100.0)	332 (99.7)
Grade 3 or higher	128 (43.5)	170 (51.1)
Serious	73 (24.8)	86 (25.8)
Led to discontinuation	45 (15.3)	51 (15.3)
Fatal	35 (11.9)	33 (9.9)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.16 Overview of adverse events by ECOG performance status

Category	0	1	2
Any event	296 (100.0)	272 (99.6)	58 (100.0)
Grade 3 or higher	148 (50.0)	129 (47.3)	21 (36.2)
Serious	84 (28.4)	65 (23.8)	10 (17.2)
Led to discontinuation	48 (16.2)	41 (15.0)	7 (12.1)
Fatal	35 (11.8)	28 (10.3)	5 (8.6)

Percentages are calculated on the number of subjects in that subgroup.

14.4 Pharmacokinetic data

Table 14.4.1 Summary of Steady-state Pharmacokinetic Parameters – Pharmacokinetic Analysis Set

Treatment	n	Cmax (ng/mL) Geometric mean (%CV)	tmax (h) Median (min, max)	AUC0-12 (ng·h/mL) Geometric mean (%CV)	t1/2 (h) Mean (SD)	CL/F (L/h) Geometric mean (%CV)
Kelvaparib 300 mg	418	1762 (29.4)	2.0 (2.0, 2.0)	17605 (36.0)	14.4 (6.82)	17.04 (36.0)

AUC0-12 = area under the concentration-time curve over one dosing interval; CL/F = apparent oral clearance; Cmax = maximum observed concentration; %CV = geometric coefficient of variation; t1/2 = terminal elimination half-life; tmax = time to maximum concentration.

Table 14.4.2 Accumulation Ratio and Trough Concentration – Pharmacokinetic Analysis Set

Treatment	n	Accumulation ratio Geometric mean (%CV)	Ctrough (ng/mL) Geometric mean (%CV)
Kelvaparib 300 mg	418	2.84 (25.6)	1079.0 (49.7)

Accumulation ratio is the ratio of AUC0-24 at steady state to AUC0-24 after the first dose.

15 REFERENCE LIST

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16 APPENDICES

16.1.1	Protocol and protocol amendments
16.1.2	Sample case report form
16.1.3	List of independent ethics committees and institutional review boards
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16.4	Individual subject data listings (archival): not applicable; individual data are provided as the analysis datasets